

Smart Room Occupancy Monitoring from Sensor Streams Using Clustering

As mentioned in our previous reports, our project aims to implement a real-time streaming pipeline for smart room occupancy detection using environmental sensor data and clustering algorithms. We have built this system on the UCI Occupancy Detection Dataset and tested our clustering algorithm on over 8000 records with features such as temperature, humidity, light, CO2, and humidity ratio. To simulate an environment similar to our use case, we use the dataset and replay it sequentially in timestamp order with a sliding window of 30 datapoints. Each record is then processed as if it were arriving from real-time sensors.

To analyze our data stream, we started by implementing a K-means algorithm and compared the predicted assignment to the ground truth labels. We initialized the model with a cold start so that it would learn clusters solely through the stream. The model has two centroids, occupied and unoccupied.

Our results show that the k-means ($k=2$) model accurately predicts the correct label with an accuracy of over 95% in occupancy classification. The cluster centroids were defined as follows:

Cluster 0 (Occupied): 2024 records - 1713 occupied, 311 unoccupied
Centroid: Temperature=21.735, Humidity=26.537, Light=440.043, CO2=993.539, HumidityRatio=0.004

Cluster 1 (Unoccupied): 6119 records - 16 occupied, 6103 unoccupied
Centroid: Temperature=20.250, Humidity=25.465, Light=13.499, CO2=478.540, HumidityRatio=0.004

In our data, the most discerning features are light (440.043 vs 13.499) and CO2 (993.539 vs 478.540). The clustering model was very strong at classifying unoccupied rooms with a false positive rate of 0.2%, but struggled a little more with occupied rooms with a false positive rate of 15%. This could likely be because of the nature of some of the data points, such as CO2 concentration and Temperature in rooms, which often takes time to dissipate, leaving residual data.

Moving forward, we plan on implementing the DBSCAN clustering algorithm, as well as more advanced analysis metrics such as precision, recall, and F1 score, along with further post-hoc analysis. We will then compare the two algorithms, K-means ($k=2$) and DBSCAN, as well as show a possible use case and application.